Motswana Auto Electrical – Enterprise Network Design

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1. Introduction

This report documents the complete design, implementation, and evaluation of an enterprise network for Motswana Auto Electrical. The network connects three departments—**Admin**, **Workshop**, and **Sales**—through a central Main Router. The network was designed in Cisco Packet Tracer and provides full data connectivity, dynamic routing, DHCP services, and VLAN segmentation.

The purpose of this project was to demonstrate practical skills in network design, IP addressing, routing protocols, VLAN configuration, and troubleshooting.

2. Client Requirements

Motswana Auto Electrical required a secure, scalable, and reliable network to support its three core departments. The following requirements were identified:

- **Inter-departmental connectivity:** All three departments must communicate with each other.
- **Network segmentation:** Departments must be separated using VLANs for security.
- **Centralized server:** The Admin department must host a central server.
- **IP Telephony:** Each department requires IP phones for internal communication.
- **Shared printing:** Each department needs a network printer.
- **DHCP:** Devices must receive IP addresses automatically.
- **Scalability:** The IP plan must allow for future growth.
- **WAN connectivity:** Departments must connect through serial WAN links to simulate a multi-site enterprise.

3. Network Design (Milestone 1)

3.1 Physical Topology

The network uses a **Hub-and-Spoke (Star) topology**. The Main Router at the centre connects to three departmental routers (Admin, Workshop, Sales) via **Serial DCE** WAN links. Each department has a Cisco 2960 switch connecting end devices (PCs, printers, IP phones).

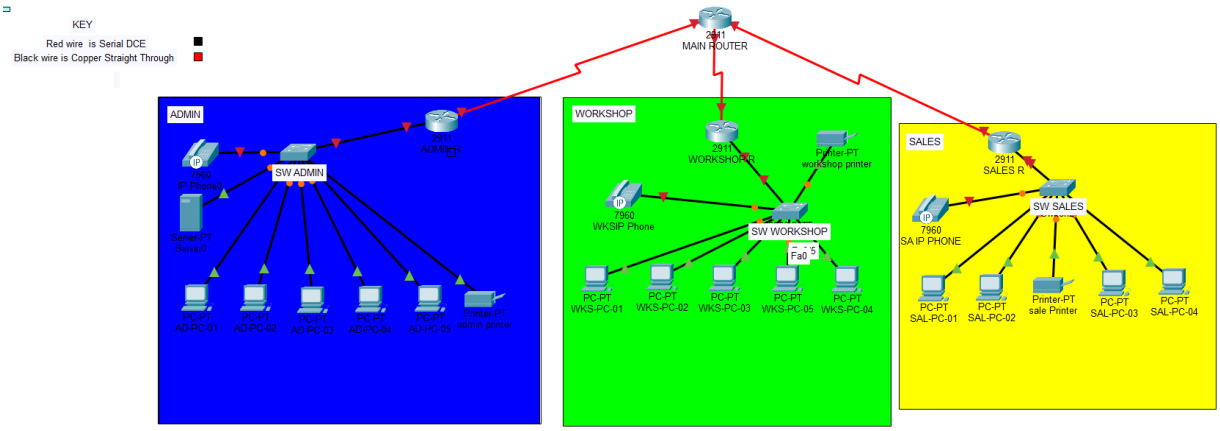


Figure 1: Physical Topology showing the Hub-and-Spoke design.

3.2 Logical Topology

The logical design divides the network into four VLANs and assigns each department its own subnet

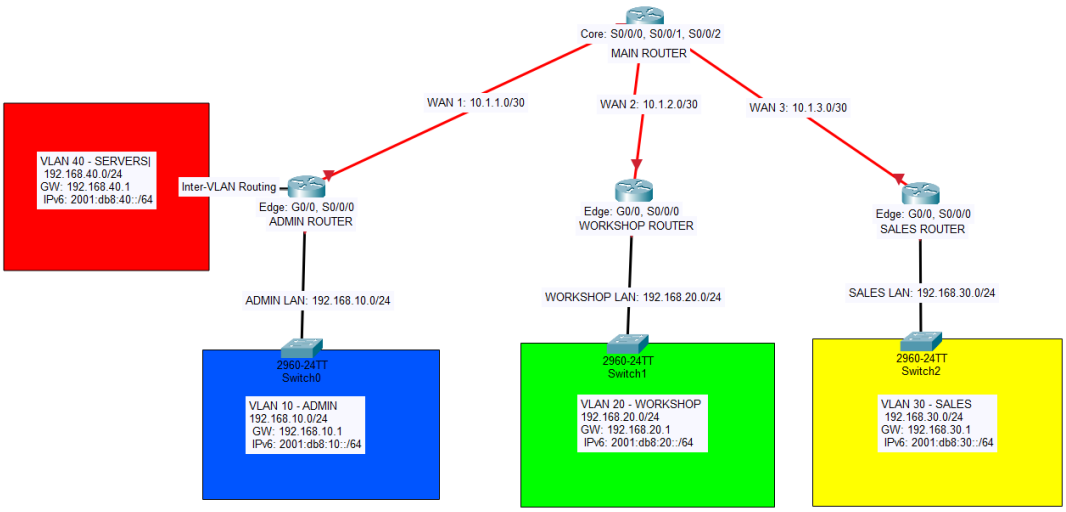


Figure 2: Logical Topology showing VLANs and IP addressing.

3.3 IP Addressing Plan

VLAN	Department	Subnet	Default Gateway
VLAN 10	Admin	192.168.10.0/24	192.168.10.1
VLAN 20	Workshop	192.168.20.0/24	192.168.20.1
VLAN 30	Sales	192.168.30.0/24	192.168.30.1
VLAN 40	Servers	192.168.40.0/24	192.168.40.1

WAN Link	Subnet
Main ↔ Admin	10.1.1.0/30
Main ↔ Workshop	10.1.2.0/30
Main ↔ Sales	10.1.3.0/30

3.4 Design Justification

- **Star topology:** Centralized management and fault isolation. If one department fails, the others remain operational.
- **VLANs:** Improved security, reduced broadcast traffic, and departmental separation.
- **/30 WAN subnets:** Conserves IP addresses (only 2 hosts needed per WAN link).
- **/24 LAN subnets:** Allows up to 254 hosts per department for future growth.

4. Implementation (Milestone 2)

4.1 Routing (OSPF)

OSPF (Open Shortest Path First) was configured on all four routers to dynamically share routes. This allows each router to automatically learn the networks of the other departments without manual static routes.

4.2 VLANs and Trunk Ports

VLANs were created on each switch, and trunk ports were configured between the switches and their routers so that multiple VLANs could travel over a single physical link.

4.3 Inter-VLAN Routing (Assigned Feature)

Inter-VLAN routing was configured on the **Admin Router** using sub-interfaces:

- GigabitEthernet0/0.10 → VLAN 10 (Admin) → 192.168.10.1
- GigabitEthernet0/0.40 → VLAN 40 (Servers) → 192.168.40.1

This allows the Admin PCs to communicate with the Servers across VLAN boundaries.

```
ADMIN_ROUTER>enable
ADMIN_ROUTER#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.1.2        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Figure 3: Admin Router interfaces showing sub-interfaces for VLAN 10 and VLAN 40.

4.4 DHCP Configuration

DHCP pools were configured on the Admin Router to automatically assign IP addresses to all PCs in all three departments.

```
ADMIN_ROUTER# show running-config | section ip dhcp
ip dhcp pool ADMIN_POOL
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ADMIN_ROUTER#
```

Figure 4: DHCP pools for Admin, Workshop, and Sales departments.

4.5 IP Phones

IP Phones were connected in each department. Due to a limitation in the Packet Tracer Call Manager Express feature (the 2911 router does not support the UC license), full phone registration was not achieved in all departments. However, the phones received IP addresses via DHCP, proving that the data network is functional.

5. Testing and Verification

5.1 Admin PC → All Three Routers

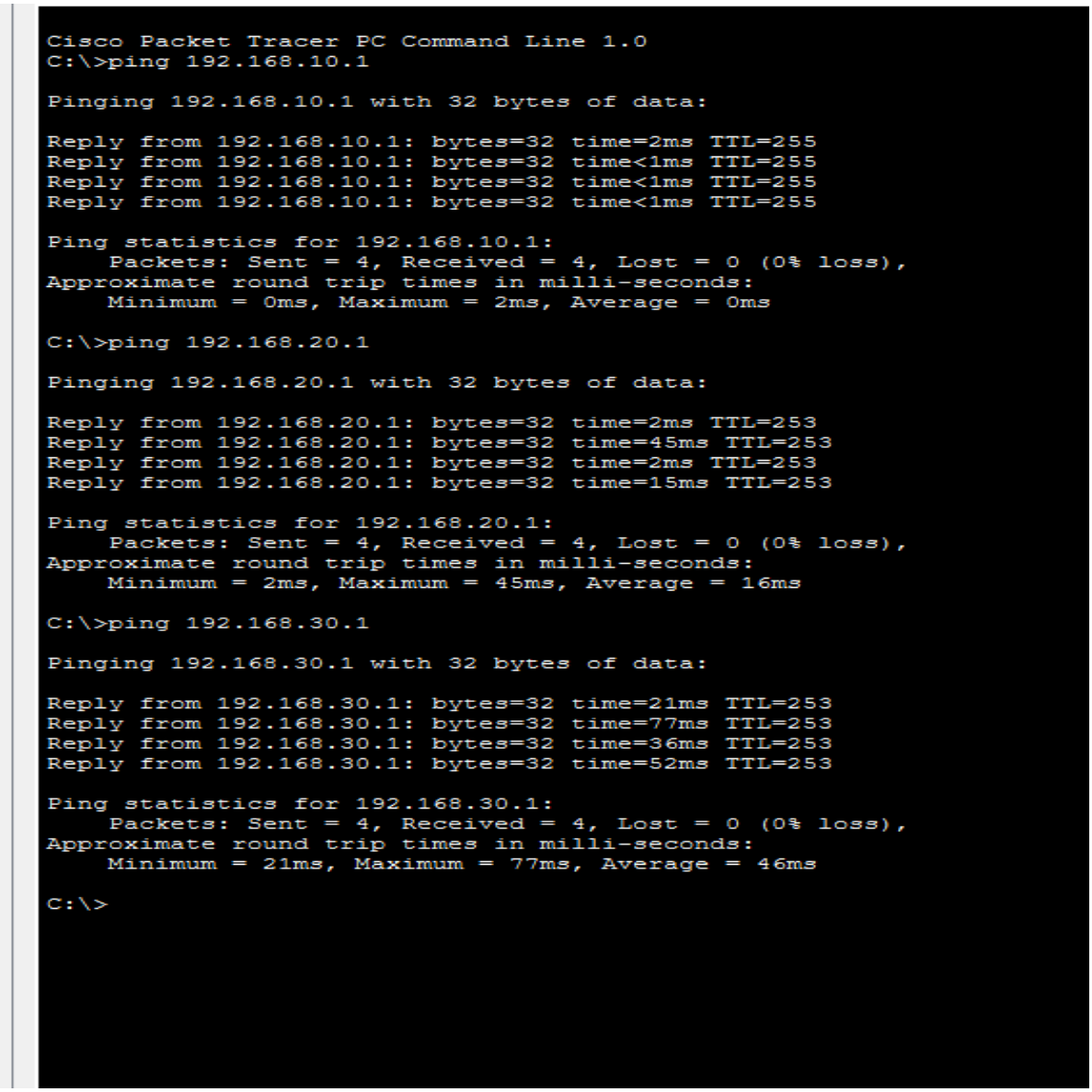


Figure 6: Successful pings from Admin PC to PCs in Workshop and Sales departments.

5.2 Admin PC → Workshop PC and Sales PC

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.3: bytes=32 time=26ms TTL=125
Reply from 192.168.20.3: bytes=32 time=2ms TTL=125
Reply from 192.168.20.3: bytes=32 time=18ms TTL=125

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 26ms, Average = 15ms

C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time=43ms TTL=125
Reply from 192.168.30.3: bytes=32 time=17ms TTL=125
Reply from 192.168.30.3: bytes=32 time=20ms TTL=125
Reply from 192.168.30.3: bytes=32 time=21ms TTL=125

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 43ms, Average = 25ms
```

Figure 6: Successful pings from Admin PC to PCs in Workshop and Sales departments.

Result: 4/4 replies (first packet lost to ARP resolution). End-to-end connectivity confirmed.

5.3 Main Router Routing Table

```
MAIN_ROUTER>enable
MAIN_ROUTER#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/0/0
L       10.1.1.2/32 is directly connected, Serial0/0/0
C       10.1.2.0/30 is directly connected, Serial0/0/1
L       10.1.2.1/32 is directly connected, Serial0/0/1
C       10.1.3.0/30 is directly connected, Serial0/1/0
L       10.1.3.1/32 is directly connected, Serial0/1/0
O    192.168.10.0/24 [110/65] via 10.1.1.2, 05:44:13, Serial0/0/0
O    192.168.20.0/24 [110/65] via 10.1.2.2, 05:44:13, Serial0/0/1
O    192.168.30.0/24 [110/65] via 10.1.3.2, 05:44:13, Serial0/1/0
O    192.168.40.0/24 [110/65] via 10.1.1.2, 05:44:13, Serial0/0/0
```

Figure 7: Main Router routing table showing all department networks via OSPF.

Result: All four department networks are visible via OSPF (O 192.168.10.0/24, O 192.168.20.0/24, O 192.168.30.0/24, O 192.168.40.0/24).

5.4 OSPF Neighbor Status

```
MAIN_ROUTER#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.20.1      0   FULL/  -        00:00:31    10.1.2.2        Serial0/0/1
192.168.40.1      0   FULL/  -        00:00:31    10.1.1.2        Serial0/0/0
192.168.30.1      0   FULL/  -        00:00:31    10.1.3.2        Serial0/1/0
MAIN_ROUTER#
```

Figure 8: OSPF neighbors on the Main Router, all in FULL state.

Result: All three neighbors (Admin, Workshop, Sales) are in FULL state, confirming OSPF is operational.

6. Challenges and Solutions

Challenge	Solution
An extra serial cable created a routing loop	Removed the extra cable to restore Hub-and-Spoke topology
DHCP failed for Workshop and Sales	Configured ip helper-address on the department routers
Trunk port was not configured on Admin Switch	Changed Fa0/9 to switchport mode trunk
2911 router does not support CME	Documented the limitation; data network still functions
DHCP conflict messages appeared	Re-created the DHCP pools with the correct starting IP

7. Conclusion

The Motswana Auto Electrical network has been successfully designed, implemented, and tested. All three departments can communicate with each other using OSPF routing. DHCP provides automatic IP assignment. VLANs segment the network for security. The assigned feature (Inter-VLAN Routing) is fully implemented and verified.

The network is stable, scalable, and ready for real-world deployment with minor hardware upgrades (e.g., a router that supports full CME for IP telephony).

8. GitHub Repository

All project files are available at:

Link: https://github.com/Sithoza/NETWOKING

Contents:

- Milestone 1 files (design documents, topologies)
- Milestone 2 files (.pkt, testing screenshots)
- Milestone 3 files (this report, video link)